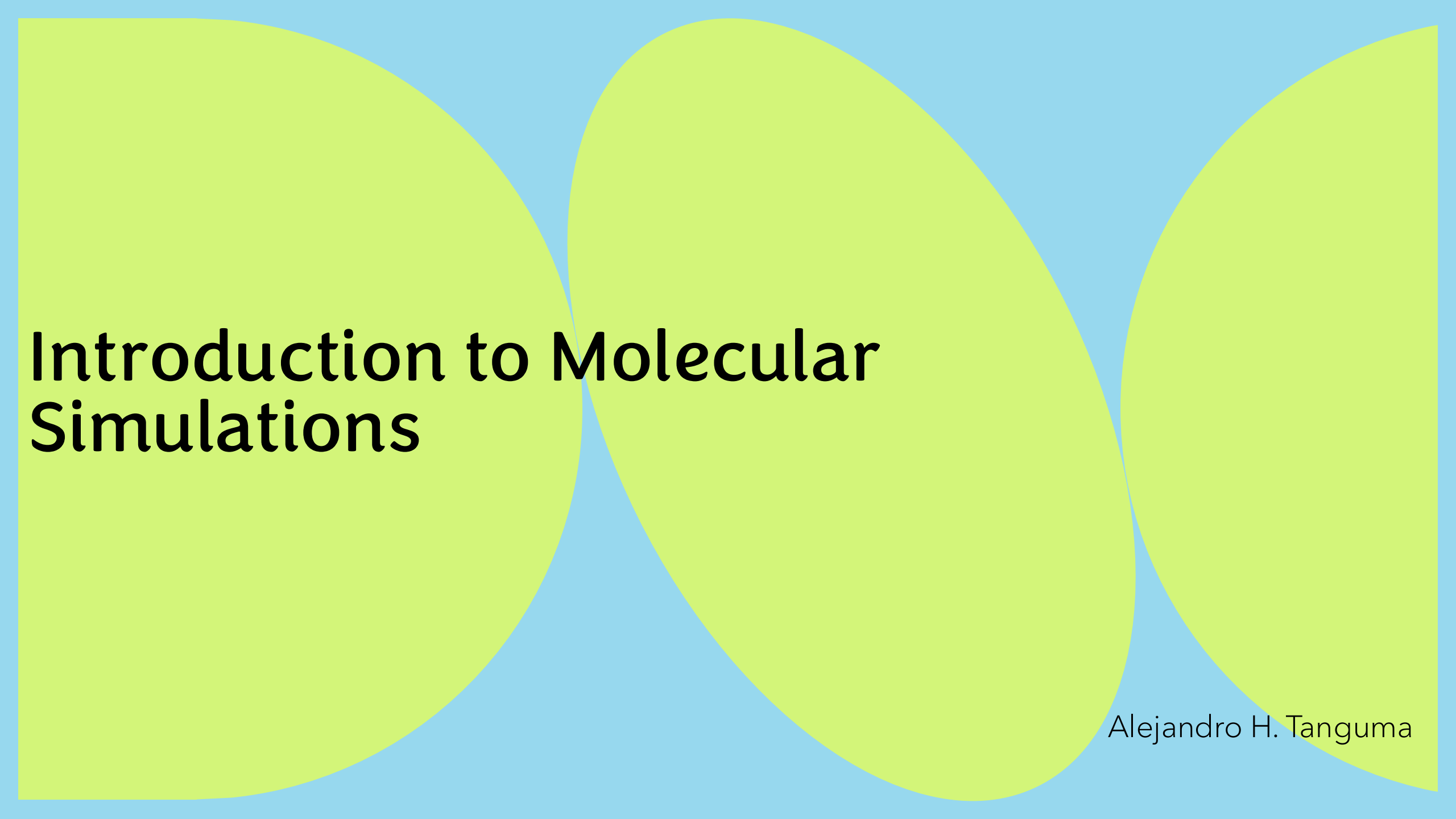




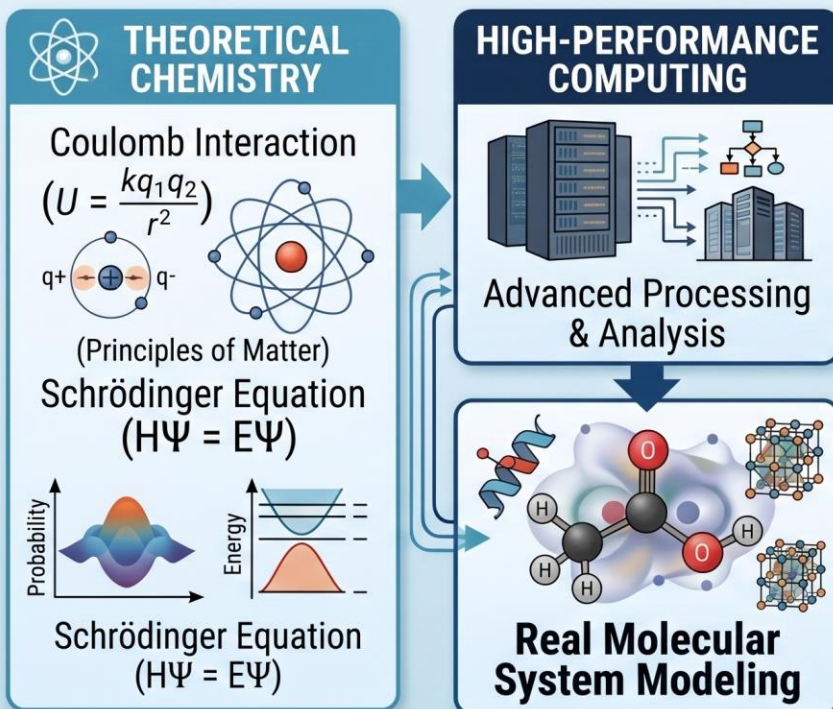
# Introduction to Molecular Simulations

Alejandro H. Tanguma

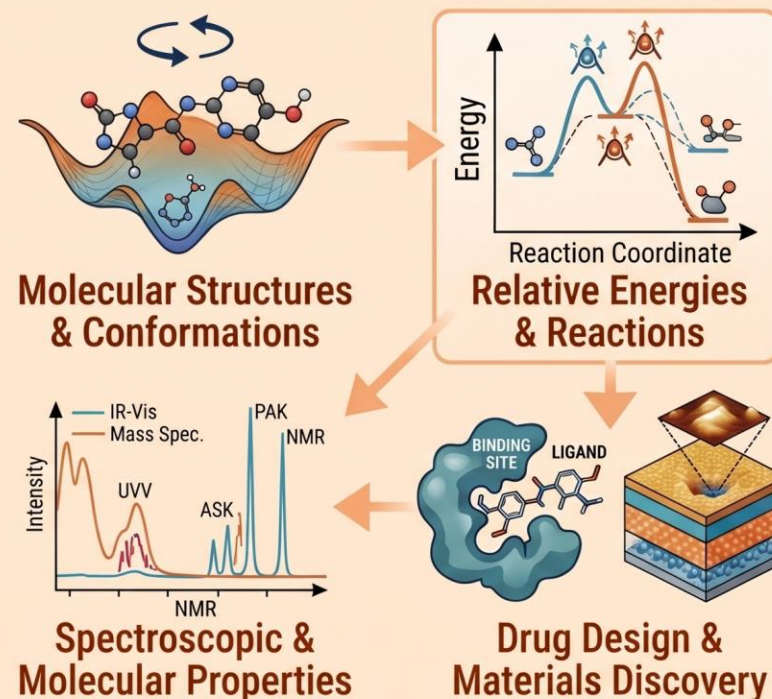
# What is computational chemistry

## WHAT IS COMPUTATIONAL CHEMISTRY & WHAT IS IT USED FOR?

### WHAT IT IS

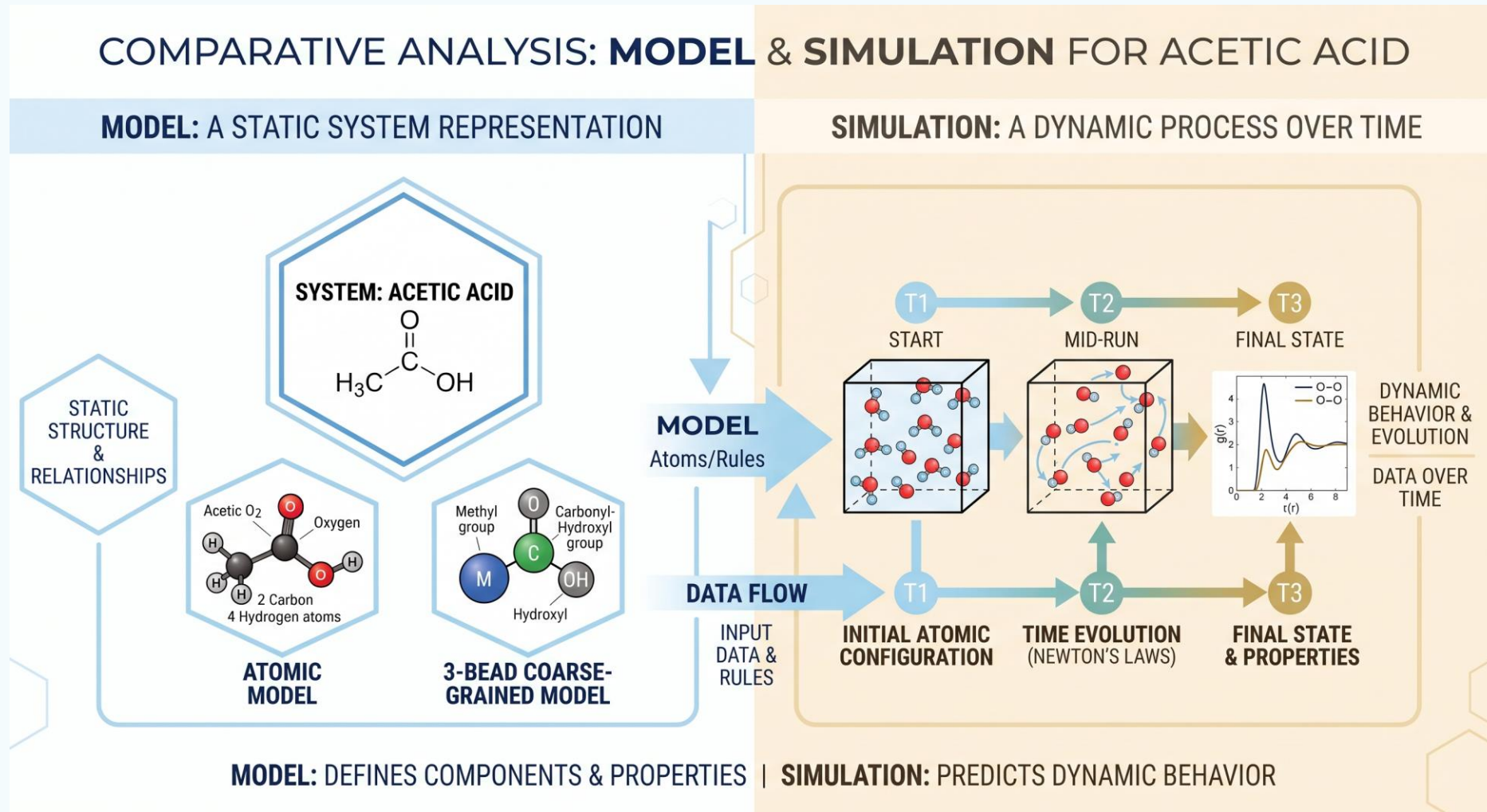


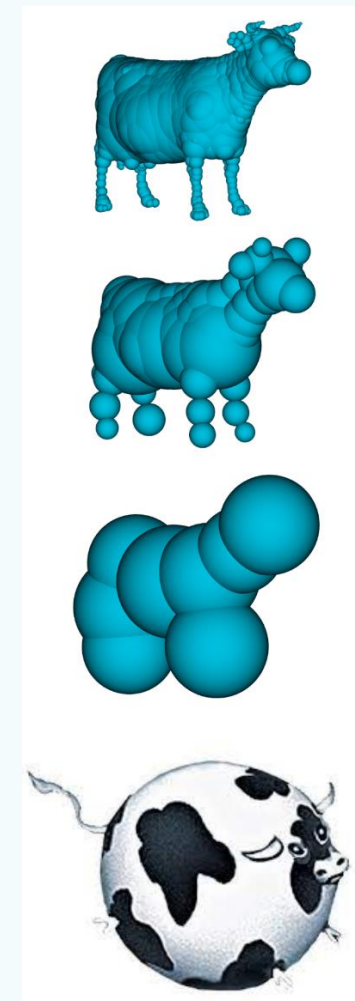
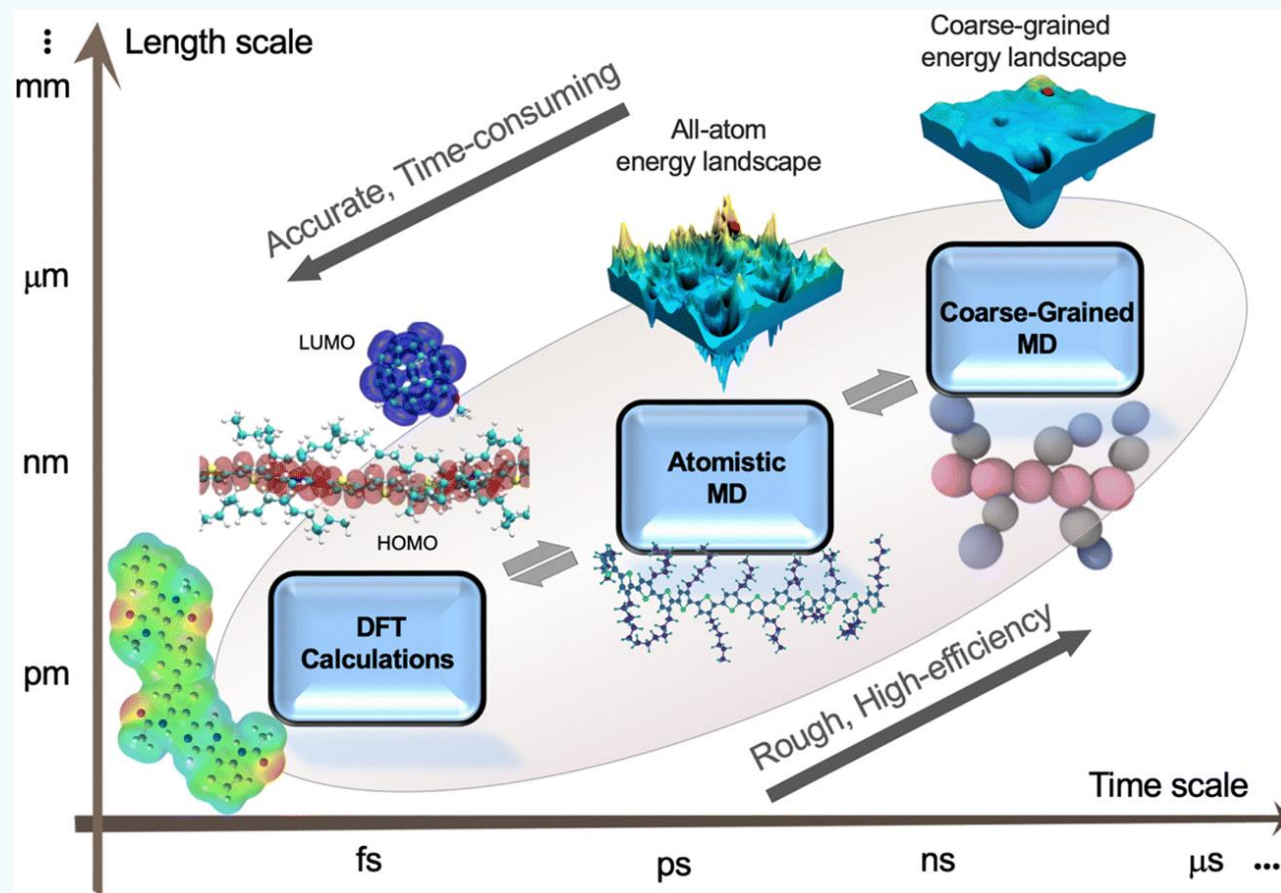
### APPLICATIONS & PREDICTIONS



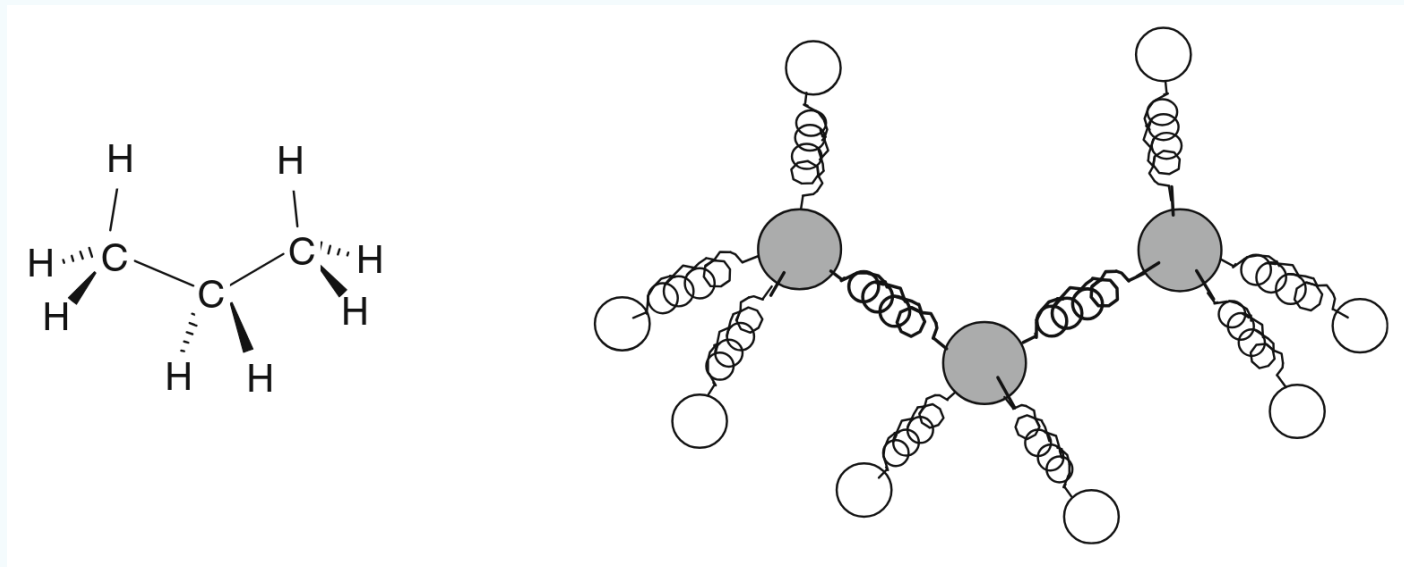
Bridge Between Theory & Experiment: Simulating Real Molecular Behavior

# Models and simulations

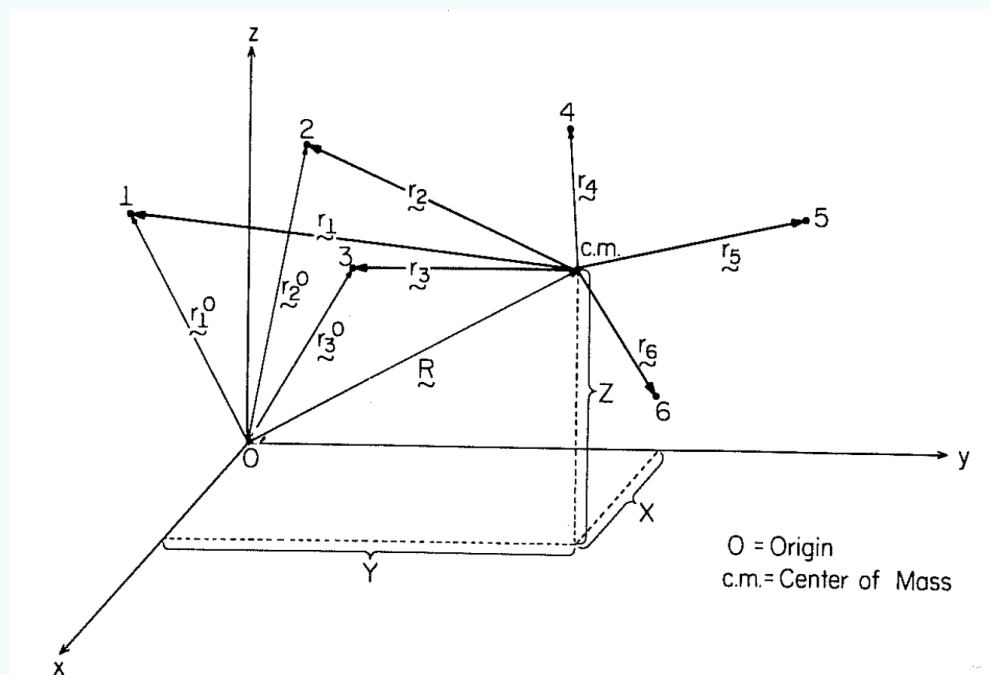




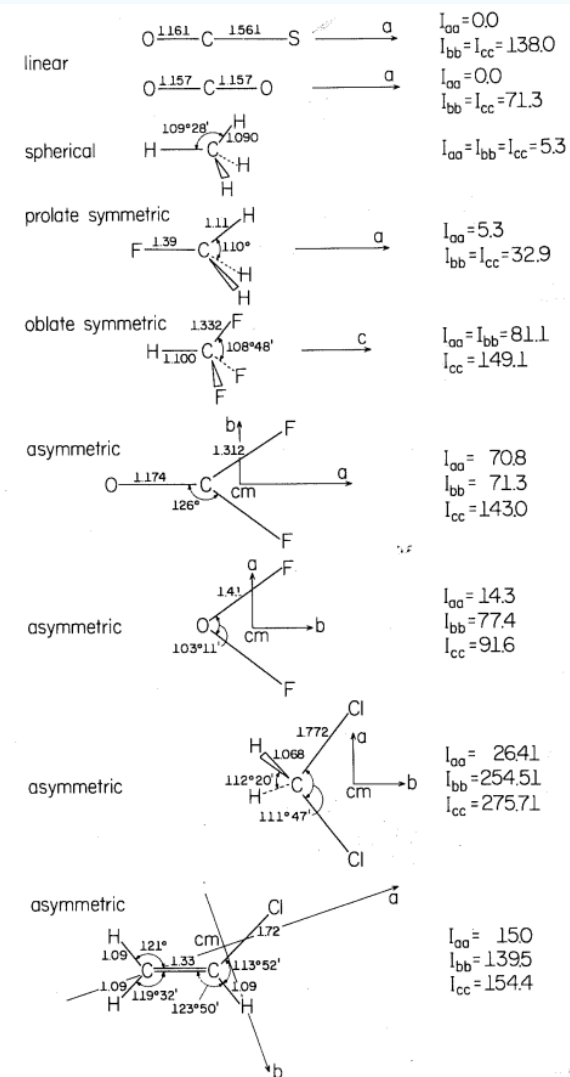
# Molecular mechanics







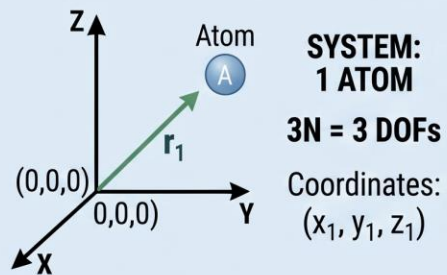
**Figure 2-1** Cartesian coordinate axis system.  $O$  is the origin of the coordinate system chosen by convenience to be the laboratory frame and c.m. is the center of mass based in the many-particle frame. The  $r_i^0$  and  $r_i$  vectors originate at the arbitrary origin and c.m., respectively, and terminate at the point masses designated by 1, 2, ....  $\mathbf{R}$  is the vector from the arbitrary origin to the c.m. and  $X$ ,  $Y$ , and  $Z$  are the Cartesian components of  $\mathbf{R}$ .



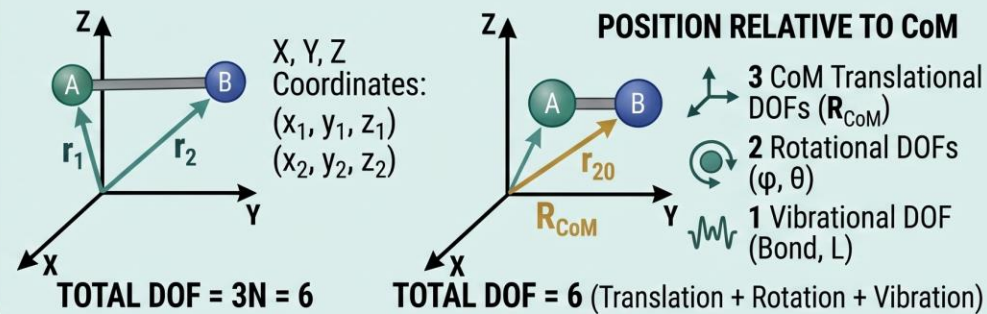
**Figure 2-2** Moments of inertia of several molecules in the principal inertial axis system ( $a$ ,  $b$ ,  $c$ ) for the common isotopes ( $^{16}\text{O}$ ,  $^{12}\text{C}$ ,  $^{32}\text{S}$ ,  $^{19}\text{F}$ , and  $^{35}\text{Cl}$ ). The bond distances are in units of  $\text{\AA} = 10^{-8} \text{ cm}$ . The units of the moments are  $10^{-40} \text{ g} \cdot \text{cm}^2$ , or  $10^{-47} \text{ kg} \cdot \text{m}^2$ .

# DEGREES OF FREEDOM (DOF) & PARTICLE COORDINATES

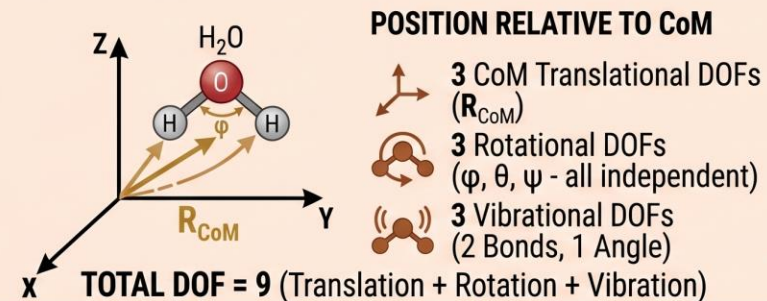
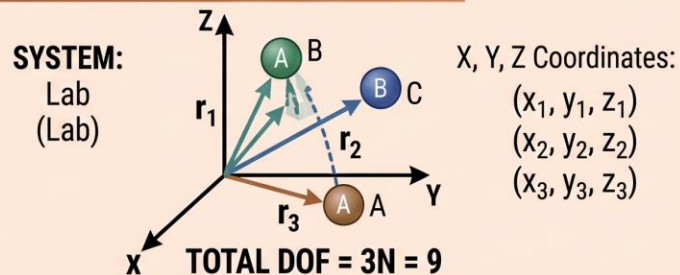
## 1 PARTICLE SYSTEM (N=1)



## 2 PARTICLE SYSTEM (N=2)

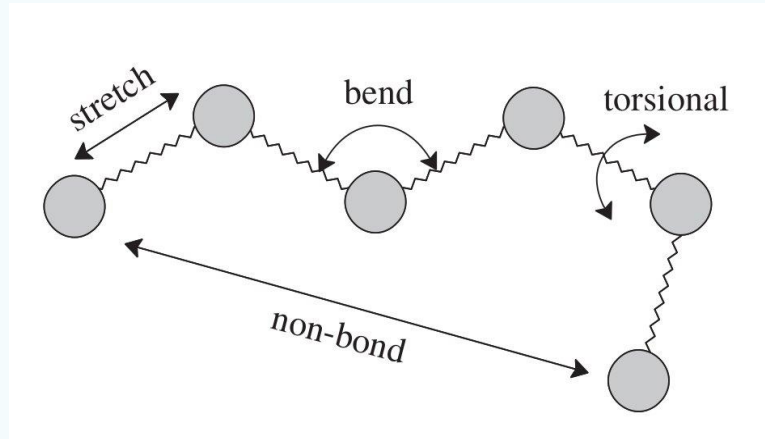


## 3 PARTICLE SYSTEM (N=3)

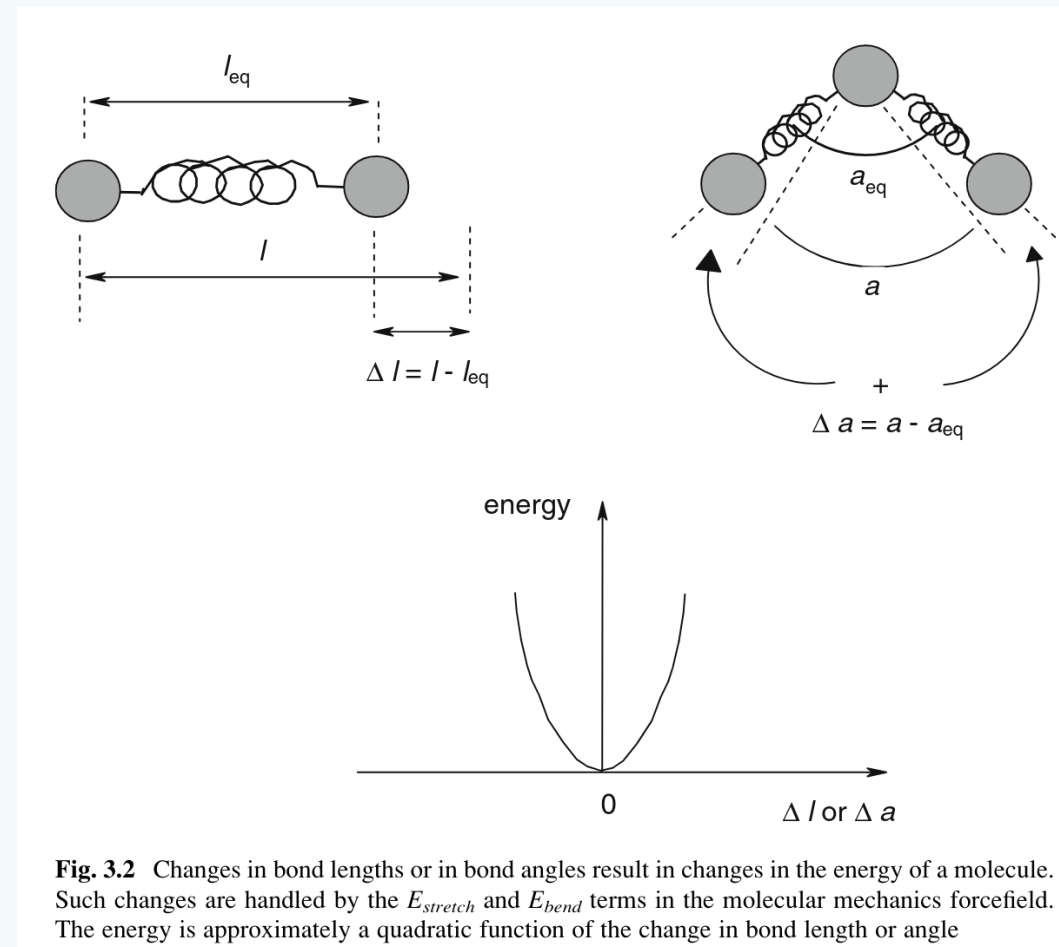
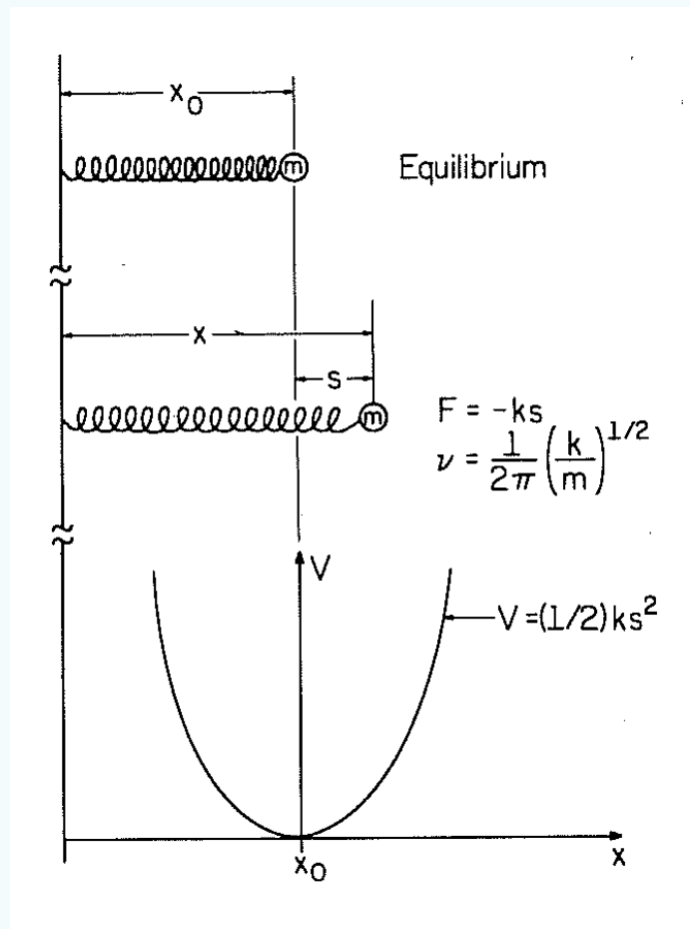


[illegible]

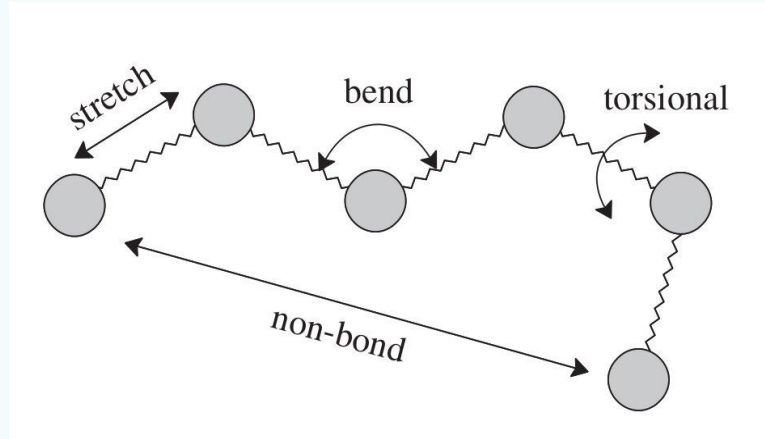




$$E = \sum_{bonds} E_{stretch} + \sum_{angles} E_{bend} + \sum_{dihedrals} E_{torsion} + \sum_{pairs} E_{nonbond}$$



**Fig. 3.2** Changes in bond lengths or in bond angles result in changes in the energy of a molecule. Such changes are handled by the  $E_{stretch}$  and  $E_{bend}$  terms in the molecular mechanics forcefield. The energy is approximately a quadratic function of the change in bond length or angle



$$\begin{aligned}
 \mathcal{V} = & \sum_{\text{bonds}} \frac{1}{2} k_r (r_{ij} - r_0)^2 + \sum_{\text{angles}} \frac{1}{2} k_\theta (\theta_{ijk} - \theta_0)^2 \\
 & + \sum_{\text{torsions}} \sum_n k_{\phi,n} [\cos(n\phi_{ijk\ell} + \delta_n) + 1] + \sum_{\text{non-bonded pairs}} \left[ \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} + \frac{A_{ij}}{r_{ij}^{12}} - \frac{B_{ij}}{r_{ij}^6} \right].
 \end{aligned}$$